

WEEK-1

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Step 1 Database Setup

Query 1: Create the database

The screenshot shows a database query editor window titled "Query 1". The query text is: `CREATE DATABASE IF NOT EXISTS bookflow_db;`. The output pane at the bottom shows a single row of results:

#	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected

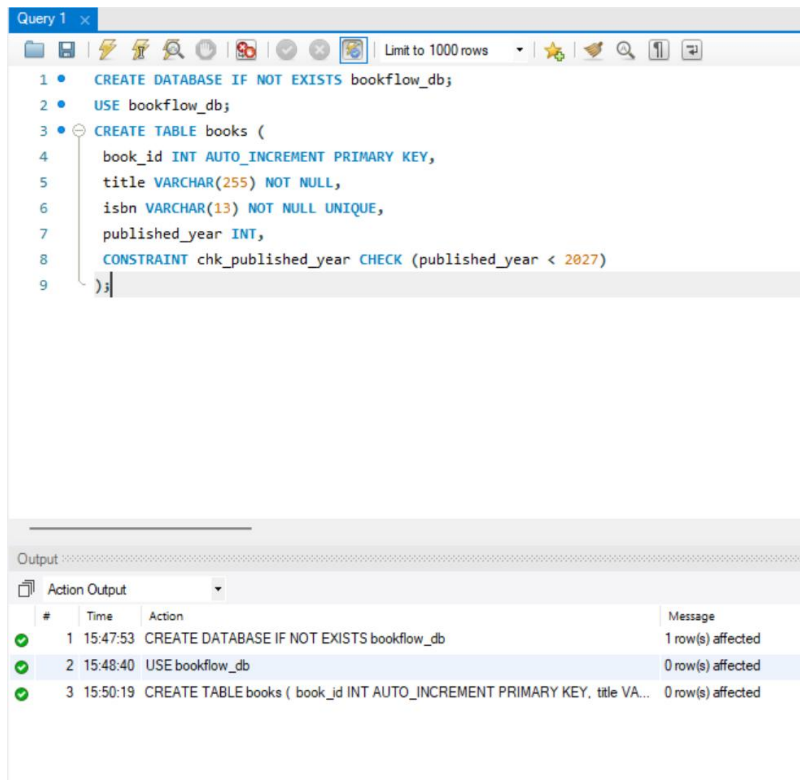
Query 2: Select the database

The screenshot shows a database query editor window titled "Query 1". The query text is: `CREATE DATABASE IF NOT EXISTS bookflow_db;` followed by `USE bookflow_db;`. The output pane at the bottom shows two rows of results:

#	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
2	15:48:40	USE bookflow_db	0 row(s) affected

Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table



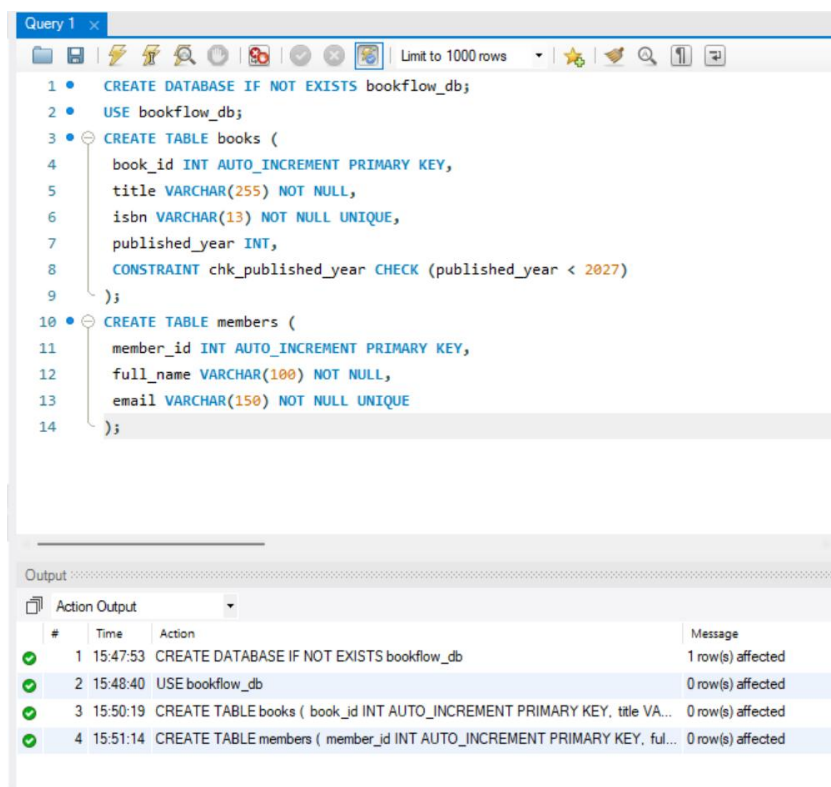
The screenshot shows a SQL IDE window titled 'Query 1'. The SQL code defines a database 'bookflow_db', uses it, and creates a table 'books' with the following specifications:

- `book_id` INT AUTO_INCREMENT PRIMARY KEY,
- `title` VARCHAR(255) NOT NULL,
- `isbn` VARCHAR(13) NOT NULL UNIQUE,
- `published_year` INT,
- `CONSTRAINT chk_published_year CHECK (published_year < 2027)`

The output pane at the bottom shows the execution results:

#	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
2	15:48:40	USE bookflow_db	0 row(s) affected
3	15:50:19	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VA...	0 row(s) affected

Query 4: Create the members table



The screenshot shows the same SQL IDE window with additional code to create a 'members' table:

- `member_id` INT AUTO_INCREMENT PRIMARY KEY,
- `full_name` VARCHAR(100) NOT NULL,
- `email` VARCHAR(150) NOT NULL UNIQUE

The output pane now shows four rows of execution results:

#	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
2	15:48:40	USE bookflow_db	0 row(s) affected
3	15:50:19	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VA...	0 row(s) affected
4	15:51:14	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, ful...	0 row(s) affected

Query 5 (verification): Inspect the table structures

Query 1

```
10 CREATE TABLE members (  
11     member_id INT AUTO_INCREMENT PRIMARY KEY,  
12     full_name VARCHAR(100) NOT NULL,  
13     email VARCHAR(150) NOT NULL UNIQUE  
14 );  
15 DESCRIBE members;
```

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 1

Output

Action Output

#	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
2	15:48:40	USE bookflow_db	0 row(s) affected
3	15:50:19	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title V...	0 row(s) affected
4	15:51:14	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, f...	0 row(s) affected
5	15:52:04	DESCRIBE members	3 row(s) returned

Step 3 Initial Seeding (Valid Data)

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES  
(  
'The Alchemist', '9780061122415', 1988),  
(  
'Clean Code', '9780132350884', 2008),  
(  
'Atomic Habits', '9780735211292', 2018);
```

Context Help Snippets

tion Output

	Time	Action	Message
1	15:47:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
2	15:48:40	USE bookflow_db	0 row(s) affected
3	15:50:19	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title V...	0 row(s) affected
4	15:51:14	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, f...	0 row(s) affected
5	15:52:04	DESCRIBE members	3 row(s) returned
6	15:53:14	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Query 7: Insert 3 members

```
40 • CREATE TABLE members (  
41     member_id INT AUTO_INCREMENT PRIMARY KEY,  
42     full_name VARCHAR(100) NOT NULL,  
43     email VARCHAR(150) NOT NULL UNIQUE  
44 );  
45 • INSERT INTO members (full_name, email) VALUES  
46 ('Manasvi', 'manasvi@gmail.com'),  
47 ('Aswitha', 'aswitha@gmail.com'),  
48 ('Tejaswini', 'tejaswini@gmail.com');  
49  
50 • INSERT INTO members (full_name, email)  
51 VALUES ('Manasvi Clone', 'manasvi@gmail.com');
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✗	11 15:59:48	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999...	Error Code: 1048. Column 'title' cannot be null
✗	12 16:00:36	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '888888...	Error Code: 3819. Check constraint 'chk_published_year' is violated.
✓	13 16:02:23	drop table members	0 row(s) affected
✗	14 16:03:25	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi@gmail.co...	Error Code: 1146. Table 'bookflow_db.members' doesn't exist
✓	15 16:03:51	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, ...	0 row(s) affected
✓	16 16:03:56	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi@gmail.co...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
53 • SELECT * FROM books;
```

Result Grid



Filter Rows:

Edit:



Export/Import:



Wrap Cell C



Result Grid



Form Editor

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

```
53 • SELECT * FROM books;
```

```
54 • SELECT * FROM members;
```

Result Grid



Filter Rows:

Edit:



Export/Import:



Wrap Cell C



Result Grid



Form Editor

	member_id	full_name	email
▶	1	Manasvi	manasvi@gmail.com
	2	Aswitha	aswitha@gmail.com
	3	Tejaswini	tejaswini@gmail.com
*	NULL	NULL	NULL

Step 4 Constraint Tests

Test 1: Duplicate ISBN (UNIQUE violation)

```
30 • INSERT INTO books (title, isbn, published_year)
31 VALUES ('Fake Copy', '9780061122415', 2000);
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 5	15:52:04	DESCRIBE members	3 row(s) returned
✓ 6	15:53:14	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	15:55:05	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi.com'), ('As...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	15:56:30	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 9	15:57:10	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 10	15:58:36	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '97800611...	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Test 2: NULL title (NOT NULL violation)

```
33 • INSERT INTO books (title, isbn, published_year)
34 VALUES (NULL, '9999999999999', 2010);
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 6	15:53:14	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	15:55:05	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi.com'), ('As...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	15:56:30	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 9	15:57:10	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 10	15:58:36	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '97800611...	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
✗ 11	15:59:48	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999...	Error Code: 1048. Column 'title' cannot be null

Test 3: Future publication year (CHECK violation)

```
36 • INSERT INTO books (title, isbn, published_year)
37 VALUES ('Time Traveler', '888888888888', 2030);
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 7	15:55:05	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi.com'), ('As...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	15:56:30	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 9	15:57:10	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 10	15:58:36	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '97800611...	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
✗ 11	15:59:48	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999...	Error Code: 1048. Column 'title' cannot be null
✗ 12	16:00:36	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '888888...	Error Code: 3819. Check constraint 'chk_published_year' is violated.

Test 4: Duplicate email (UNIQUE violation)

```
50 • INSERT INTO members (full_name, email)
51 VALUES ('Manasvi Clone', 'manasvi@gmail.com');
```

[Context Help](#) [Snippets](#)

Output

Action Output				
#	Time	Action	Message	
✖	12 16:00:36	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '888888...	Error Code: 3819. Check constraint 'chk_published_year' is violated.	
✔	13 16:02:23	drop table members	0 row(s) affected	
✖	14 16:03:25	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi@gmail.co...	Error Code: 1146. Table 'bookflow_db.members' doesn't exist	
✔	15 16:03:51	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, ...	0 row(s) affected	
✔	16 16:03:56	INSERT INTO members (full_name, email) VALUES ('Manasvi', 'manasvi@gmail.co...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	
✖	17 16:04:42	INSERT INTO members (full_name, email) VALUES ('Manasvi Clone', 'manasvi@g...	Error Code: 1062. Duplicate entry 'manasvi@gmail.com' for key 'members.email'	